

Hands-On-Learning: Responsible AI Design Challenge

Learning Objectives

By the end of this hands-on practice, you will be able to:

- Map ethical risks across the data, modeling, and deployment stages of the AI lifecycle
- Use a Responsible AI Checklist to document fairness, accountability, and transparency safeguards
- Develop a practical Bias Mitigation Plan with actionable techniques
- Reflect on how ethical considerations shape technical and communication decisions



Description

In this hands-on Visual Studio Code practice, you'll act as a Responsible AI designer at InsightMart Analytics, a company developing an AI model to predict customer purchasing behavior. The business team is eager to launch quickly, but you must ensure the design aligns with Responsible AI principles such as fairness, accountability, and transparency. You will document ethical risks across the AI lifecycle and create a bias-mitigation plan that balances innovation speed with responsible design.

This lab focuses on applying Responsible AI principles to real-world system design. You'll analyze where risks can arise—during data collection, model training, and deployment—and propose concrete safeguards. Using a simplified checklist and bias-mitigation plan, you'll practice embedding fairness and transparency requirements directly into your project documentation. By the end, you'll understand how to design AI systems that deliver value while respecting ethics and governance expectations.

Structure of the Activity

Dataset Information

File 1: responsible_ai_checklist.json – structured JSON for listing risks, principles, and mitigations across lifecycle stages

File 2: bias_mitigation_plan.yaml – YAML template for summarizing bias-reduction strategies

Key Features: Templates support three AI lifecycle stages (Data Collection, Modeling, Deployment) and core RAI principles (Fairness, Accountability, Transparency)

Tools to be used:

- Visual Studio Code (any recent version)
- Python 3.10 or higher
- No external libraries required (uses built-in json and yaml-compatible parsing)

Setup Requirements

- **Technical Prerequisites:** Python 3.10+ installed and accessible from command line
- **System Requirements:** Any operating system with Visual Studio Code support
- **Required Files in Working Directory:** responsible_ai_checklist.json, bias_mitigation_plan.yaml, reflection.md (all created during the lab)

Scenario

You are a Responsible AI designer at InsightMart Analytics, a company developing an AI model to predict customer purchasing behavior. The business team is eager to launch within weeks, but leadership has requested a design review before moving to model training. Your task is to identify ethical risks at each stage of the AI lifecycle, propose mitigations, and create a bias-reduction plan that satisfies both innovation goals and governance requirements.

Objective

Complete a Responsible AI Checklist documenting risks across all lifecycle stages, develop a Bias Mitigation Plan with at least three actionable strategies, and write a reflection on integrating ethical considerations into AI design.



Getting Started

- 1 Open Visual Studio Code and create a new folder for your lab work
- 2 Create a new file named `responsible_ai_checklist.json` with a JSON structure containing three lifecycle stages
- 3 Create a new file named `bias_mitigation_plan.yaml` for your mitigation strategies
- 4 Create a new file named `reflection.md` for your written reflection
- 5 Verify Python is installed by running `python --version` in the terminal

Instructions

Activity 1: Identify Lifecycle Risks

Objective

Map ethical risks across the three AI lifecycle stages (Data Collection, Modeling, Deployment) and document them in a structured JSON checklist.



Input/Data

Your responsible_ai_checklist.json file (initially empty or with basic structure)

Context: InsightMart's AI model uses customer transaction history, demographic data, and browsing patterns to predict purchasing behavior

Steps to Complete

Step 1: Design a JSON structure with three main sections representing AI lifecycle stages: "data_collection", "modeling", and "deployment". Each section should support an array of risk entries.

Step 2: For the Data Collection stage, identify at least one ethical risk (e.g., biased sampling, privacy concerns in data gathering, lack of consent documentation). Assign the risk to the most relevant Responsible AI principle.

Step 3: For the Modeling stage, identify at least one ethical risk (e.g., algorithmic bias, lack of explainability, overfitting to majority groups). Map to the appropriate RAI principle.

Step 4: For the Deployment stage, identify at least one ethical risk (e.g., model drift affecting fairness, lack of human oversight, inadequate error handling). Assign to the relevant principle.

Step 5: Write 1–2 sentences describing a realistic mitigation for each identified risk.

Expected Output

A valid JSON file with three lifecycle stage sections

At least one documented risk per stage (minimum 3 total)

Each risk entry includes: risk_id, description, principle, and mitigation fields

Tips

Consider how customer demographics might create bias at each stage

Use VS Code's JSON validation to catch syntax errors early

Think about both technical risks (model behavior) and process risks (governance gaps)

Activity 2: Develop a Bias Mitigation Plan

Objective

Create a structured Bias Mitigation Plan in YAML format documenting at least three concrete methods to reduce bias and improve transparency.



Input/Data

Your bias_mitigation_plan.yaml file

Insights from your lifecycle risk analysis in Activity 1

Steps to Complete

Step 1: Design a YAML structure that organizes mitigation strategies with fields for: strategy name, description, lifecycle stage it addresses, and expected impact.

Step 2: Document your first mitigation strategy addressing data-level bias. Consider techniques such as balanced data sampling, stratified collection, or synthetic data augmentation for underrepresented groups.

Step 3: Document a second strategy focused on model transparency. Consider techniques such as explainability tools (SHAP, LIME), fairness audits, or interpretable model architectures.

Step 4: Document a third strategy for ongoing monitoring. Consider post-deployment fairness metrics, model drift detection, or regular bias testing across demographic groups.

Step 5: For each strategy, write a brief explanation of how it reduces bias or improves transparency in the InsightMart context.

Expected Output

A valid YAML file with structured mitigation strategies

At least three distinct mitigation methods documented

Each strategy includes name, description, stage, and expected impact

Tips

YAML uses indentation (spaces, not tabs) to define structure—be consistent

Focus on actionable, measurable strategies rather than vague goals

Consider how each strategy connects to the specific risks you identified in Activity 1

Activity 3: Reflect on Responsible Design Integration

Objective

Write a reflection on how Responsible AI principles were integrated into your design process and stakeholder communication.



Input/Data

Your reflection.md file

Completed work from Activities 1 and 2

Steps to Complete

Step 1: Open your reflection.md file and create a heading for your reflection

Step 2: Write 150–200 words addressing how you integrated Responsible AI principles into the design documentation

Step 3: Discuss how the checklist and mitigation plan could be communicated to different stakeholders (technical team, business leadership, compliance)

Step 4: Reflect on the balance between speed-to-market pressures and responsible design requirements

Expected Output

A markdown file with 150–200 words of reflective writing

Clear connection between documentation artifacts and stakeholder communication

Acknowledgment of the tension between innovation speed and ethical design

Tips

Consider how the same risk might be explained differently to engineers versus executives

Think about how documentation enables accountability and auditability

Reference specific examples from your checklist and mitigation plan

Compile Your Findings

Requirement 1: Responsible AI Checklist

Completed responsible_ai_checklist.json file

Include:

- Three lifecycle stage sections (Data Collection, Modeling, Deployment)
- At least one risk documented per stage
- Each risk mapped to appropriate Responsible AI principle
- Mitigation strategy for each identified risk
- Valid JSON syntax

Requirement 2: Bias Mitigation Plan

Completed bias_mitigation_plan.yaml file

Include:

- At least three distinct mitigation strategies
- Strategy name and description for each
- Lifecycle stage addressed by each strategy
- Expected impact on bias reduction or transparency
- Valid YAML syntax

Requirement 3: Design Integration Reflection

Completed reflection.md file

Include:

- 150–200 words of reflective analysis
- Discussion of stakeholder communication approaches
- Reflection on balancing innovation speed with responsible design

Summary

This activity teaches Responsible AI design documentation through practical implementation in a realistic retail analytics scenario. You've learned how to systematically identify risks across the AI lifecycle, create actionable mitigation plans, and communicate ethical considerations to stakeholders—skills directly applicable in AI product management, data science, and governance roles where balancing innovation with responsibility is essential.

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.